Sample Allocation and Computer-Assisted Sampling

A Technical Guidebook for Excel and Stata Implementation

Dr. Anil Shrestha

National Statistical Training Reference Guide

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Session Outline

Why Allocation Matters: The Problem We Are Solving

The Core Question

In a stratified sample survey, after segmenting the population into H mutually exclusive strata (e.g., provinces, urban/rural districts), how do we distribute our total sample budget of n units across strata (n_h)?

- **Statistical Precision:** Minimizes standard errors of national estimators.
- **Budget Efficiency:** Allocates resources strategically depending on field operational costs.
- **Analytical Validity:** Ensures small sub-groups (domains) are adequately represented.

The Fallacy of Simple Equal Division

The Naive Approach:

- Dividing $n = 3,000$ equally across Nepal's 7 provinces (≈ 429 per province).
- Ignores massive population disparities.
- Results in highly volatile **sampling fractions** (f_h) and unbalanced **design weights** (w_h).

Table: Table 1: Nepal Households vs. Equal Allocation

Province	Pop. (N_h)	Equal n_h	Sampling Fraction
Koshi	720k	429	0.060%
Madhesh	810k	429	0.053%
Bagmati	960k	429	0.045%
Gandaki	360k	429	0.119%
Lumbini	630k	429	0.068%
Karnali	180k	429	0.238%
Sudurpashchim	240k	429	0.179%
Total	**3.9M**	**3,003**	**0.077%**

Analytical Variables Controlled by Allocation

An allocation plan directly determines the primary variables used in complex survey analysis:

1. Sampling Fraction (f_h)

The probability of selecting a unit within stratum h:

$$f_h = \frac{n_h}{N_h}$$

2. Design Weights (w_h)

The expansion factor used to scale up sample statistics to the population:

$$w_h = \frac{1}{f_h} = \frac{N_h}{n_h}$$

Warning

Extreme variation in design weights drastically inflates the **Design Effect (DEFF)** and standard errors of national estimators.

The Three Pillars of Stratified Allocation

- **Population Size (N_h):** Larger strata require larger sample sizes to keep sampling fractions balanced.
- **Internal Variability (S_h):** High internal diversity requires larger samples. Homogeneous strata are cheap to estimate precisely.
- **Sampling Unit Cost (c_h):** Remote, high-cost sectors should be sampled less when variance targets permit.

Table: Table 2: Statistical Vulnerabilities of Poor Sample Allocation

Design Error	Consequence
Oversampling large strata	Wastes budget; suppresses small-domain representation.
Undersampling small strata	Unstable domain estimates for key administration zones.
Ignoring variance (S_h)	High national aggregate standard errors.
Ignoring operational cost (c_h)	Triggers severe fieldwork budget overruns.
Absence of sample floors	Stratum $n_h \leq 1$ makes variance estimation impossible.

The Running Reference Dataset

We will use this dataset representing a national **Household Income Survey** throughout the guidebook.

- Total population (*N*): $1,900,000$ households across $H = 5$ strata.
- Total contract sample size (*n*): $1,200$ households.

Table: Table 3: Baseline Strata Parameters for the Running Reference Dataset

Stratum (h)	**Geographical Description**	**Population (N_h)**	**Income Std. Dev. (S_h, NPR '000)**
1	Mountain Rural	80,000	12.0
2	Hill Rural	340,000	15.0
3	Hill Urban	260,000	22.0
4	Terai Rural	780,000	18.0
5	Terai Urban	440,000	28.0
Total		**1,900,000**	

Equal vs. Proportional Allocation Methods

Equal Allocation

Distributes sample size uniformly:

$$n_h = \frac{n}{H}$$

- In our case: $n_h = 240$ units.
- **When to use:** Highly exploratory designs, pilot studies, or when domain comparison is the primary goal.
- **Drawback:** Severe weight variations (from $w_1 = 333$ to $w_4 = 3,250$).

Proportional Allocation

Based strictly on population share:

$$n_h = n \times \frac{N_h}{N}$$

- In our case: $n_4 \approx 1,200 \times \frac{780\text{k}}{1.9\text{M}} \approx 493$.
- **When to use:** Standard baseline for self-weighting (EPSEM) designs.
- **Drawback:** Blind to internal stratum variation (S_h) and operational costs (c_h).

Comparison of Equal and Proportional Allocations

Table: Table 4: Equal vs. Proportional Sample Allocation Comparison

Stratum	Description	Equal n_h	Proportional n_h	Net Difference
1	Mountain Rural	240	51	-189
2	Hill Rural	240	215	-25
3	Hill Urban	240	164	-76
4	Terai Rural	240	493	+253
5	Terai Urban	240	277	+37
Total		**1,200**	**1,200**	

- Under Proportional, Mountain Rural drops to only 51 households. This is too small for reliable sub-national estimates.
- Under Equal, high-density hubs like Terai Rural are under-sampled (240 vs. 493), causing severe national variance inflation.

Neyman (Optimal) Sample Allocation

Developed by Jerzy Neyman (1934), this method incorporates population size and internal variability simultaneously to minimize the variance of the overall population estimator $V(\bar{y}_{st})$ for a fixed total sample size n.

The Neyman Formula

$$n_h = n \times \frac{N_h S_h}{\sum_{h=1}^{H} N_h S_h}$$

Efficiency Gain Metrics

$$\text{Efficiency Gain} = \frac{V_{\text{proportional}}(\bar{y}_{st})}{V_{\text{Neyman}}(\bar{y}_{st})}$$

An efficiency gain of 1.15 indicates Neyman achieves the same precision as a proportional sample that is 15% larger!

Neyman Allocation: Step-by-Step Workflow

Step 1: Compute Priority Products ($N_h \times S_h$)

Table: Table 5: Priority Value Calculations for Neyman Optimal Allocation

Stratum	Description	Population (N_h)	Std. Dev. (S_h)	Product ($N_h S_h$)
1	Mountain Rural	80,000	12.0	960,000
2	Hill Rural	340,000	15.0	5,100,000
3	Hill Urban	260,000	22.0	5,720,000
4	Terai Rural	780,000	18.0	14,040,000
5	Terai Urban	440,000	28.0	12,320,000
Sum				**38,140,000**

Step 2 & 3: Calculate Ratios, Multiply by $n = 1,200$, and Resolve Rounding

- Stratum 4 Ratio: $\frac{14,040,000}{38,140,000} \approx 0.36809$ $\Longrightarrow$ Raw $n_4 \approx 441.71$ $\Longrightarrow$ Rounded $n_4 = 442$
- Sum of rounded values = $1,201$. We subtract 1 unit from Stratum 2 (rounded from 160.46 to 161 with 0.46 remainder) to restore the budget to exactly $1,200$.

Comparison: Equal, Proportional, and Neyman

Table: Table 6: Comparison of Equal, Proportional, and Neyman Allocations

Stratum	Description	Pop. (N_h)	SD (S_h)	Equal	Proportional	Neyman
1	Mountain Rural	80,000	12.0	240	51	30
2	Hill Rural	340,000	15.0	240	215	160
3	Hill Urban	260,000	22.0	240	164	180
4	Terai Rural	780,000	18.0	240	493	442
5	Terai Urban	440,000	28.0	240	277	388
Total		**1,900,000**		**1,200**	**1,200**	**1,200**

Key Insights into Neyman's Behavior:

- **Terai Urban ($S_h = 28.0$):** Extremely high internal variance leads to an increased allocation of 388 (up from proportional's 277).
- **Mountain Rural ($S_h = 12.0$):** Very homogeneous; Neyman reduces its sample to just 30.
- **Hill Urban vs. Hill Rural:** Hill Urban has smaller population but higher variance ($22.0 > 15.0$), thus receiving a larger sample size ($180 > 160$).

Power Allocation and Square Root Variant

Proportional allocation leaves small regions with too few samples, whereas equal division compromises national estimates. **Power Allocation** uses a tuning exponent $q \in [0, 1]$ to control this balance.

Power Allocation Formula

$$n_h = n \times \frac{N_h^q}{\sum_{h=1}^{H} N_h^q}$$

The Dial of the Exponent (q):

- $q = 1.0$: Restores Proportional.
- $q = 0.0$: Restores Equal.
- $q = 0.5$: **Square Root Allocation** (Standard in international surveys).

Table: Table 7: Square Root (Power $q = 0.5$) Allocation Calculations

Stratum	Description	Sqrt $\sqrt{N_h}$	Final n_h
1	Mountain Rural	282.84	116
2	Hill Rural	583.10	239
3	Hill Urban	509.90	209
4	Terai Rural	883.18	363
5	Terai Urban	663.32	273
Total		**2,922.34**	**1,200**

Cost-Constrained Optimal Allocation

Neyman allocation assumes that the unit cost of interviewing is uniform. In practice, interviewing in mountainous, remote rural areas can cost several times more than in dense urban corridors.

Cost-Constrained Optimal Allocation Formula

$$n_h = n \times \frac{N_h S_h / \sqrt{c_h}}{\sum_{h=1}^{H} N_h S_h / \sqrt{c_h}}$$

where c_h is the unit cost of interviewing a household in stratum h.

Suppose the unit costs of interviewing in our five strata are:

- Mountain Rural: $c_1 = \text{NPR } 8,000$ (highly remote)
- Hill Rural: $c_2 = \text{NPR } 4,000$
- Hill Urban: $c_3 = \text{NPR } 3,000$
- Terai Rural: $c_4 = \text{NPR } 2,500$
- Terai Urban: $c_5 = \text{NPR } 2,000$ (highly accessible)

Cost-Constrained Optimal Allocation: Results

Table: Table 8: Cost-Constrained Optimal Allocation Calculations

Stratum	Description	Product ($N_h S_h$)	Unit Cost (c_h)	Factor ($N_h S_h / \sqrt{c_h}$)	Final n_h
1	Mountain Rural	960,000	8,000	10,733	**17**
2	Hill Rural	5,100,000	4,000	80,664	129
3	Hill Urban	5,720,000	3,000	104,451	167
4	Terai Rural	14,040,000	2,500	280,800	448
5	Terai Urban	12,320,000	2,000	275,486	439
Total				**752,134**	**1,200**

Design Risk

The sample size for Mountain Rural falls to only 17 households. This is statistically invalid for calculating stable standard errors or performing sub-group analysis. We must apply a **minimum sample floor**!

Minimum Sample Size Floors & Top-Up Algorithm

To prevent zero- or near-zero allocations, we enforce a minimum sample floor (n_{min}), typically ranging from 30 to 50 depending on the survey type.

The Multi-Stage "Top-Up and Re-Allocate" Algorithm

1. If any stratum falls below the floor n_{min}, lock its sample size to exactly n_{min}.
2. Calculate the remaining sample budget:

$$n_{\text{remaining}} = n - \sum n_{\text{min}}$$

3. Re-allocate the remaining sample only among unconstrained strata using the original formula.
4. Repeat check iteratively if necessary.

Applying $n_{\text{min}} = 30$ to our Cost-Constrained results:

- Mountain Rural: $17 < 30 \implies$ locked to 30.
- Remaining sample to re-allocate: $1,200 - 30 = 1,170$ units distributed across Strata 2 to 5.

Comparison of All Major Allocation Methods

Table: Table 9: Multi-Stage Iterative Flooring and Top-Up Results ($n_{\text{min}} = 30$)

Stratum	Description	Equal	Proportional	Neyman	Cost-Constrained (Floor 30)
1	Mountain Rural	240	51	30	**30**
2	Hill Rural	240	215	160	127
3	Hill Urban	240	164	180	165
4	Terai Rural	240	493	442	443
5	Terai Urban	240	277	388	435
Total		**1,200**	**1,200**	**1,200**	**1,200**

Key Takeaway

The floor-adjusted cost-constrained design successfully safeguards Mountain Rural's regional estimation capacity at 30, with the 13-unit top-up burden distributed across the rest of the country.

Excel Spreadsheet Architecture

To build a dynamic, interactive sample allocation sheet, place your global parameter input blocks in a separate area (e.g., Columns Q and R) to separate logic from raw data.

Column Header Layout (Rows 1-7, Columns A-P):

Col	Variable / Formula Concept
A	Stratum ID
B	Geographical Description
C	Population (N_h)
D	Standard Deviation (S_h)
E	Unit Cost (c_h)
F	Equal Allocation (n_h)
G	Population Share (W_h)
H	Proportional Allocation (n_h)
I–K	Neyman Product, Proportion, and Allocation (n_h)
L–N	Power Term (N_h^q), Power Proportion, and Power Allocation (n_h)
O–P	Cost-Constrained Product ($N_h S_h / \sqrt{c_h}$), and Allocation (n_h)

Global Settings:

- **Cell R1:** Total Sample (n) $= 1200$
- **Cell R2:** Exponent (q) $= 0.5$
- **Cell R3:** Floor (n_{min}) $= 30$

Excel Formulas (Row 2 Examples)

Enter these formulas in Row 2 and copy down through Row 6 to ensure dynamic updates:

- **Equal Allocation (Col F):**
  `=ROUND($R$1 / COUNT($C$2:$C$6), 0)`
- **Population Share (Col G) & Proportional Allocation (Col H):**
  `=C2/SUM($C$2:$C$6)` and `=ROUND($R$1 * G2, 0)`
- **Neyman Proportion (Col J) & Neyman Allocation (Col K):**
  `=I2/SUM($I$2:$I$6)` and `=ROUND($R$1 * J2, 0)` *(where I2 = C2*D2)*
- **Power Term (Col L) & Power Allocation (Col N):**
  `=C2^$R$2` and `=ROUND($R$1 * (L2/SUM($L$2:$L$6)), 0)`
- **Cost-Constrained Allocation (Col P):**
  `=ROUND($R$1 * (O2/SUM($O$2:$O$6)), 0)` *(where O2 = (C2*D2)/SQRT(E2))*

Column Total Checks (Row 7)

Verify that all allocations sum to exactly *n* by entering: `=SUM(F2:F6)`, `=SUM(H2:H6)`, etc.

Stata Implementation: Initialization & Baseline Data

Step 1: Input Baseline Data and Globals

```
clear all
set seed 20830310     // Set seed for reproducibility

input str20 description double N_h double S_h double c_h
"Mountain Rural" 80000 12 8000
"Hill Rural"     340000 15 4000
"Hill Urban"     260000 22 3000
"Terai Rural"    780000 18 2500
"Terai Urban"    440000 28 2000
end
gen stratum = _n

global n_total = 1200    // Target sample size
global q_power = 0.5     // Power exponent
global n_min = 30        // Minimum sample floor
```

Stata Script: Classical, Neyman & Power Allocations

Steps 2 & 3: Standard Allocation Formulas

```
* Equal and Proportional
quietly count
local H = r(N)
gen n_equal = round($n_total / `H', 1)

quietly summarize N_h
local N_total = r(sum)
gen W_h = N_h / `N_total'
gen n_prop = round($n_total * W_h, 1)

* Neyman Allocation
gen NhSh = N_h * S_h
quietly summarize NhSh
local sum_NhSh = r(sum)
gen n_neyman = round($n_total * (NhSh / `sum_NhSh'), 1)

* Power Allocation
gen Nh_q = N_h ^ $q_power
quietly summarize Nh_q
```

Stata Script: Dynamic Minimum Flooring Loop

Step 5: The "Top-Up and Re-Allocate" Iterative Loop

```
gen n_floor = $n_total * (NhSh / `sum_NhSh')   // Start with raw Neyman
local converged = 0
local iter = 0

while `converged' == 0 {
    local iter = `iter' + 1
    gen byte is_floored = (n_floor < $n_min)
    quietly summarize is_floored
    local num_floored = r(sum)
    local allocated_to_floored = `num_floored' * $n_min
    local n_remaining = $n_total - `allocated_to_floored'

    quietly summarize NhSh if is_floored == 0
    local sum_NhSh_free = r(sum)

    replace n_floor = $n_min if is_floored == 1
    replace n_floor = `n_remaining' * (NhSh / `sum_NhSh_free') if is_floored == 0
    drop is_floored

    quietly count if n_floor < $n_min
    if r(N) == 0 {
        local converged = 1
    }
}
gen n_neyman_floor = round(n_floor, 1)
```

Stata Script: Drawing Sample & Declaring Design

Steps 6 & 7: Sample Selection and Design Declaration

```
* Simulate a population frame of 1,900 clusters/units for demonstration
set obs 1900
gen hh_id = _n
gen stratum = 1 + (hh_id>80) + (hh_id>420) + (hh_id>680) + (hh_id>1460)

* Merge the final allocations back to frame and draw sample
merge m:1 stratum using "allocation_results.dta", nogenerate
gen rand_draw = runiform()
bysort stratum (rand_draw): gen selection_rank = _n
gen hh_selected = (selection_rank <= n_allocated)

* Set populations and calculate design weights
gen N_h_real = 80000 if stratum == 1
replace N_h_real = 340000 if stratum == 2
replace N_h_real = 260000 if stratum == 3
replace N_h_real = 780000 if stratum == 4
replace N_h_real = 440000 if stratum == 5

gen weight = N_h_real / n_allocated

* Declare the complex survey design
svyset hh_id [pweight=weight], strata(stratum)
svydescribe
```

Rounding Drift and the Largest Remainder Method

Because strata sample allocations (n_h) must be whole numbers, rounding can cause the sum to drift away from the target total (n), leaving the final sample size slightly over or under budget.

The Largest Remainder Solution

1. Round all raw stratum sample sizes (n_h) **down** to the nearest integer.
2. Calculate the budget shortfall: Shortfall $= n - \sum \lfloor n_h \rfloor$.
3. Sort strata by their fractional remainders: $n_h - \lfloor n_h \rfloor$ in descending order.
4. Add exactly 1 unit to each of the top strata until the shortfall is fully resolved.

Table: Table 10: Rounding Drift and Largest Remainder Adjustment

Stratum Description	**Raw n_h**	**Rounded Down**	**Remainder**	**Final Adjusted n_h**
Mountain Rural	30.20	30	0.20	30
Hill Rural	160.46	160	0.46	160
Hill Urban	180.00	180	0.00	180
Terai Rural	441.71	441	0.71	$441 + 1 = 442$
Terai Urban	387.60	387	0.60	$387 + 1 = 388$
Total	**1,200.00**	**1,198**	**(Shortfall = 2)**	**1,200**

Practical Field Adjustments: Over & Under Allocation

Over-Allocation ($n_h > N_h$)

- Occurs in small strata with extremely high variance.
- **Solution:** Designate as a **"Take-All" Stratum**, and set $n_h = N_h$.
- Remove the stratum from the sampling design, convert it to a census segment, and re-allocate the remaining budget to other strata.

Under-Allocation ($n_h \leq 1$)

- Strata with small populations and low variance may get close to 0 or 1 sample.
- **Statistical Failure:** $n_h = 0$ causes complete bias. $n_h = 1$ makes stratum standard error calculations impossible.
- **Solution:** Enforce a strict minimum floor ($n_{\text{min}} \geq 30$).

Kish's Weight Dispersion Approximation

Unequal probability designs generate variance inflation. We can approximate the **Design Effect (DEFF)** caused by weight dispersion using Kish's formula:

Kish's DEFF Formula

$$\text{DEFF}_{\text{weights}} \approx 1 + \text{CV}(w_h)^2$$

where $\text{CV}(w_h)$ is the coefficient of variation of the design weights across the selected sample.

Operational Guide

A $\text{DEFF}_{\text{weights}}$ value above **2.0** indicates that the allocation introduces too much variance. If this occurs, researchers should adjust the allocation back towards a proportional layout.

Final Sample Design Weights and Kish's DEFF Calculations

Table: Table 11: Final Sample Design Weights Analysis

Stratum	Pop. (N_h)	Sample (n_h)	Weight (w_h)	Factor (N_h^2/n_h)
1. Mountain Rural	80,000	30	2,666.67	2.13×10^8
2. Hill Rural	340,000	127	2,677.17	9.10×10^8
3. Hill Urban	260,000	165	1,575.76	4.10×10^8
4. Terai Rural	780,000	443	1,760.72	13.73×10^8
5. Terai Urban	440,000	435	1,011.49	4.45×10^8
Total	**1.90M**	**1,200**	**—**	**3.35×10^9**

Kish's Dispersion Calculation:

Formula

$$\text{DEFF}_{\text{weights}} = \frac{n \sum_{h=1}^{H} \frac{N_h^2}{n_h}}{N^2}$$

Substitution:

$$\text{DEFF}_{\text{weights}} = \frac{1,200 \times 3.352 \times 10^9}{(1.90 \times 10^6)^2} = \frac{4.022 \times 10^{12}}{3.610 \times 10^{12}} \approx \mathbf{1.114}$$

Interpretation: Weight dispersion causes a minor 11.4% variance inflation, which is safely below the standard design warning limit of 2.0.

Final Pre-Fieldwork Quality Checks

Before releasing the sample allocation plan to field teams, verify that it passes these critical checks:

- ☐ **Budget Alignment:** The sum of allocated samples matches the national total exactly ($\sum n_h = n$).
- ☐ **Minimum Floors Met:** Every stratum has at least n_{min} (typically ≥ 30) to support variance estimation.
- ☐ **Over-Allocation Check:** Confirm that no stratum has $n_h > N_h$.
- ☐ **Weight Dispersion:** Keep the ratio of the maximum to minimum design weight reasonable (ideally < 4.0) to avoid inflating standard errors.
- ☐ **Cost Feasibility:** Ensure the projected fieldwork cost ($\sum n_h \times c_h$) remains within the approved financial budget limits.